9.3.1 Galvanic cell

We have already learnt in XI standard that when a zinc metal strip is placed in a copper sulphate solution, the blue colour of the solution fades and the copper is deposited on the zinc strip as red – brown crust due to the following spontaneous chemical reaction.

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu\ (s)$$

The energy produced in the above reaction is lost to the surroundings as heat.

In the above redox reaction, Zinc is oxidised to Zn^{2+} ions and the Cu^{2+} ions are reduced to metallic copper. The half reactions are represented as below.

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^- \quad \text{(oxidation)}$$

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu\ (s) \quad \text{(reduction)}$$

If we perform the above two half reactions separately in an apparatus as shown in figure 9.5, some of the energy produced in the reaction will be converted into electrical energy. Let us understand the function of a galvanic cell by considering Daniel cell as an example. It uses the above reaction for generation of electrical energy.

The separation of half reaction is the basis for the construction of Daniel cell. It consists of two half cells.

Oxidation half cell

A metallic zinc strip that dips into an aqueous solution of zinc sulphate taken in a beaker, as shown in Figure 9.5.

Reduction half cell

A copper strip that dips into an aqueous solution of copper sulphate taken in a beaker, as shown in Figure 9.5.

Joining the half cells

The zinc and copper strips are externally connected using a wire through a switch (k) and a load (example: volt meter). The electrolytic solution present in the cathodic and anodic compartment are connected using an inverted U tube containing agar-agar gel mixed with an inert electrolytes such as KCl, Na_2SO_4 etc., The ions of inert electrolyte do not react with other ions present in the half cells and they are not either oxidised (or) reduced at the electrodes. The solution in the salt bridge cannot get poured out, but through which the ions can move into (or) out of the half cells.

When the switch (k) closes the circuit, the electrons flows from zinc strip to copper strip. This is due to the following redox reactions which are taking place at the respective electrodes.

XII U9 Electro Chemistry.indd 44 11-01-2021 14:44:03